101

Peritoneal Dialysis: Principles, Techniques, and Adequacy

Joanne M. Bargman, Bourne L. Auguste

INTRODUCTION

The prevalence of peritoneal dialysis (PD) continues to grow steadily worldwide. It is estimated that more than 272,000 patients worldwide are treated with PD, accounting for about 11% of the total global dialysis population.[1] Patients on PD can have numerous fluid exchanges through a permanent indwelling silicone-based catheter. The exchange of PD fluid involves three important steps: filling, dwelling, and draining. Sterile dialysate provided in plastic bags is instilled into the peritoneal cavity with each exchange. The peritoneal membrane and surrounding capillaries function as a semipermeable membrane, which allows removal of waste and fluid. The removal of waste from the blood occurs by diffusion. Fluid is removed by a process called ultrafiltration (UF), in which an osmotic gradient generated by glucose or by another osmotic agent leads to water flux across the membrane.

PD can be prescribed up to 24 hours/day and 7 days/wk to patients in the form of continuous ambulatory peritoneal dialysis (CAPD) or continuous cyclic peritoneal dialysis (CCPD). In CCPD, patients receive treatment with an automated PD (APD) cycler, allowing for nighttime exchanges followed by a long dwell called a "last fill" that remains in the peritoneum during the daytime. There is significant geographic variation in the use of PD cyclers, driven in part by local resources and reimbursement policies. For example, in lower-resource settings, the overall PD uptake, prevalence, and cycler use are reduced compared with countries where reimbursement is greater.[2] CAPD is the preferred dialysis modality in low- and middle-income countries as it is cost-effective and does not require more expensive equipment such as automated PD cyclers.[3]

ADVANTAGES AND LIMITATIONS OF PERITONEAL DIALYSIS

PD can be offered to many patients with kidney failure barring a few absolute contraindications. These contraindications include abdominal hernias that cannot be corrected surgically, abdominal adhesions not amenable to adhesiolysis, recurrent intraabdominal infections, and large hiatal hernias. Furthermore, severe cognitive impairment is another contraindication, unless a helper or visiting health care worker is available.

Patients with residual kidney function (RKF) may have greater benefit from PD compared with anuric patients. PD prescriptions can be adjusted easily to ensure effective clearance when RKF declines. However, if adequate dialysis or UF cannot be achieved or maintained with PD, patients may need to be transferred to hemodialysis (HD).

PD offers numerous advantages to patients. First, PD can be initiated in an incremental manner, which allows incident patients to become comfortable in the early phases of their treatment while minimizing treatment burden. Second, PD allows for slow and gentle removal of fluid without significant hemodynamic changes. Third, PD can preserve RKF better than HD.[4] Fourth, PD does not require vascular access and, unlike HD, there is no blood-membrane contact, which has been suggested as a cause of accelerated decline in RKF.[2] Observational data suggests that PD can effectively remove excess fluid in patients with refractory volume overload due to cardiorenal syndrome.[5]

PD patients have better self-reported quality of life than patients receiving other types of dialysis. PD offers flexibility in schedule and travel while reducing the frequency for hospital visits. In patients receiving kidney transplants, PD is associated with better 5-year overall patient survival and a lower risk of delayed graft function compared with patients previously treated with HD.[3]

There are, however, some important disadvantages of PD that must be considered by patients, caregivers, and practitioners. PD is usually offered in the home, where the burden of care for patients and family members can be overwhelming, especially among patients who are frail or unable to safely perform PD without daily assistance. PD also increases hyperglycemia risk, particularly when using high-concentration glucose-based solutions. There is also an increased risk of dyslipidemia along with volume overload. Volume overload risk can be reduced by prescribing high-dose diuretics in patients with RKF.

PRINCIPLES OF PERITONEAL DIALYSIS

The fundamental principles of solute and fluid transport across the peritoneal membrane in PD include diffusion, osmosis, and convection. The peritoneal membrane consists of two major layers, the mesothelial monolayer and submesothelial compact zone (Fig. 101.1A). The compact zone contains peritoneal capillaries that allow for solute clearance and fluid removal. Over time, changes in the peritoneal membrane can occur, leading to neovascularization of capillaries and fibrosis, and ultimately to changes in membrane transport characteristics (Fig. 101.1B).

Three-Pore Model

Peritoneal membrane transport occurs through three distinct pores of varying sizes, in what is described as the three-pore model (Fig. 101.2). Small pores allow small-solute and fluid exchange between the plasma and peritoneal cavity through the capillary wall in spaces called interendothelial clefts. These clefts have a radius of 40 to 50 Å, preventing the movement of larger macromolecules such as immunoglobulins and other macromolecules.[6] Larger molecules can move between the plasma and peritoneal cavity via very large pores (radius ~250 Å) in the capillaries and postcapillary venules. Very large pores only account for 0.01% of the total capillary pores, and transport across these pores is mainly driven by hydrostatic pressure.[6] The smallest pore, also called aquaporin-1 (AQP-1, radius $<$ 4 Å), is

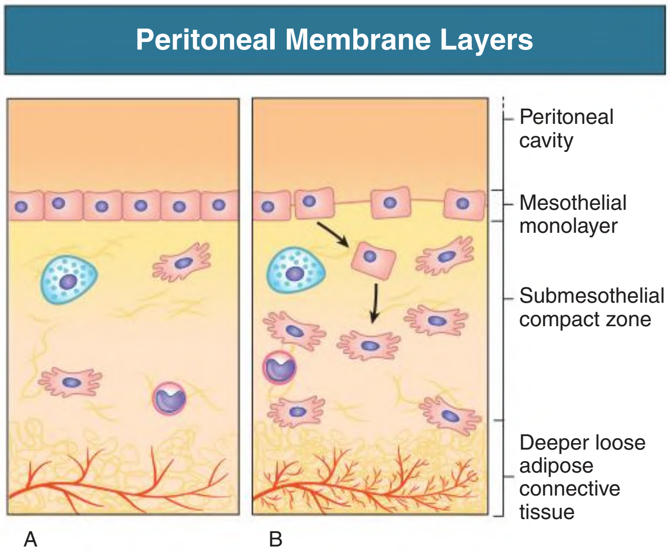

Fig. 101.1 (A) various layers of the peritoneal membrane. (B) Changes occurring over time with peritoneal dialysis in peritoneal membrane layers with the submesothelial compact zone increasing in thickness due to extracellular matrix expansion along with neovascularization of capillaries in the deep layers. (From Nessim SJ, Perl J, Bargman JM. The renin-angiotensin-aldosterone system in peritoneal dialysis: is what is good for the kidney also good for the peritoneum? *Kidney Int*. 2010;78[1]:23–28.)

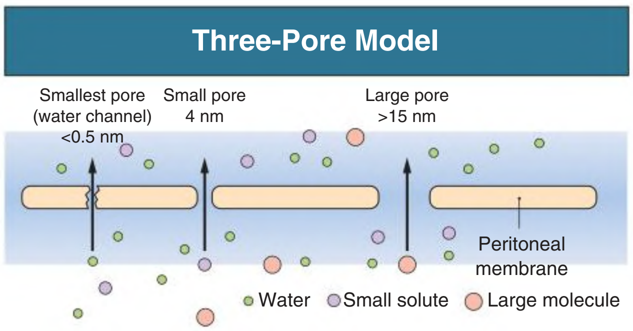

Fig. 101.2 Three-Pore Model. Small pores (4 nm) represent the major pathway across the peritoneum through which small solutes move by diffusion and water by convection driven by hydrostatic, colloid osmotic, and crystalloid osmotic pressure differences. Across large pores (>15 nm), macromolecules move out slowly by convection from plasma to the peritoneal cavity. The smallest pores (<0.5 nm) are represented by aquaporins permeable to water but are impermeable to solutes. Water moves here exclusively by crystalloid osmotic pressure.

responsible for nearly 50% of osmotically driven water movement in PD[7] (Fig. 101.1).

Effective Peritoneal Surface Area

The effective peritoneal surface area represents an area of the peritoneum that is perfused by capillaries and remains in direct contact with dialysate. Peritoneal blood flow through the capillaries can vary depending on vasodilation and vasoconstriction but is estimated to be 50 to 100 mL/min.[8] Variations in peritoneal capillary vascular resistance can also influence the permeability-surface area product (PS), also known as the mass transfer area coefficient. For example, inflammation from peritonitis can potentiate capillary vasodilation, increasing the capillary surface area and in turn enhancing small solute PS. Although small solute clearance may be enhanced with an increase in capillary recruitment, the same cannot be said about fluid permeability. Alterations in peritoneal blood flow do not directly translate to changes in peritoneal fluid permeability. The peritoneal fluid permeability is also known as hydraulic conductance (L_pS) and is dependent on the distribution and density of AQP-1 and small pores across the peritoneum.[9] L_pS varies from individual to individual such that in some cases of peritonitis, there is an increase in the L_pS, leading to increased fluid transport across all pores. The opening of large pores may increase the leakage of macromolecules such as albumin into the peritoneum from the plasma. Enhanced peritoneal absorption of fluid from the dialysate can occur with peritonitis, resulting in reduced fluid removal due to rapid dissipation of the osmotic gradient between peritoneum and plasma.

Changes in patient positioning and fill volume can have significant variation in the dialysate-peritoneal surface area. Most adult patients will tolerate a maximum fill volume of 2 to 2.5 L or an intraperitoneal hydrostatic pressure (IPP) of less than 18 cm of H_2O without much discomfort.[10] However, at higher pressures (≥18 cm of H_2O), patients normally highlight discomfort even in a supine position. IPP is highest when patients are sitting compared with standing and is lowest in the supine position. A common misconception is that IPP elevations could result in high transcapillary hydrostatic pressure gradients that may lead to an increased loss of fluid from the peritoneal cavity. However, approximately 80% of the change in IPP (ΔIPP) will transmit to large veins surrounding the peritoneal cavity, causing increased intracapillary hydrostatic pressure.[11,12] As a result, the transcapillary hydrostatic pressure gradient is reduced, and the impact of ΔIPP on fluid absorption is minimal.

Fluid Kinetics

In PD, small pores account for over 95% of all pores in the peritoneal membrane. Only 2% of all pores in the peritoneal membrane are AQP-1, but they account for about 50% of water transport that is driven by the osmotic gradient created by intraperitoneal hyperosmolar dialysate.[13] Glucose-based dialysate solutions generate water transport through both AQP-1 and small pores. On the other hand, colloid osmotic agents such as icodextrin remove fluid primarily through small pores (~90%). The movement of electrolyte-free water via AQP-1 results in a fall in dialysate concentration of sodium during the first 2 hours of the dialysate dwell, a process referred to as sodium sieving. As dwell duration extends beyond 2 hours, sodium diffuses from plasma to dialysate through the small pores along its concentration gradient, and the dialysate concentration of sodium approaches the level in plasma.

Computer models have simulated the variation in UF over a 12-hour dwell time (Fig. 101.3A). The osmotic gradient is greatest in the initial phase of the dwell, resulting in a large initial UF volume that diminishes with time as the osmotic gradient is lost. As the osmotic gradient dissipates, fluid absorption occurs through small pores (60%–80%) and lymphatics (20%–40%) in the surrounding tissues. Fig. 101.4 demonstrates the partial flows in the peritoneal membrane modeled across the various fluid conductive pathways in the three-pore model for 3.86% glucose PD solution. In "rapid transporters" (high and high-average transporters), the osmotic gradient created by glucose dissipates rapidly with UF occurring early in the dwell and fluid reabsorption in the latter part of the dwell. Conversely, "slow transporters" have prolongation of the glucose osmotic gradient, and thus better and more prolonged UF. Glucose has a low osmotic efficiency (osmotic reflection coefficient [σ] = 0.03), indicating that the osmotic gradient is more rapidly lost compared with many other solutes. On the other hand, glucose has an osmotic efficiency of 100% across AQP-1 (σ = 1).[14] As a result of this increased efficiency across AQP-1, glucose exerts about a 30-fold increase in the transport of fluid through aquaporins and leads to movement of solute-free water across the membrane. Fig. 101.3B demonstrates that for 4.25% glucose solution, the dialysate concentration of sodium will drop from 132 mmol/L to

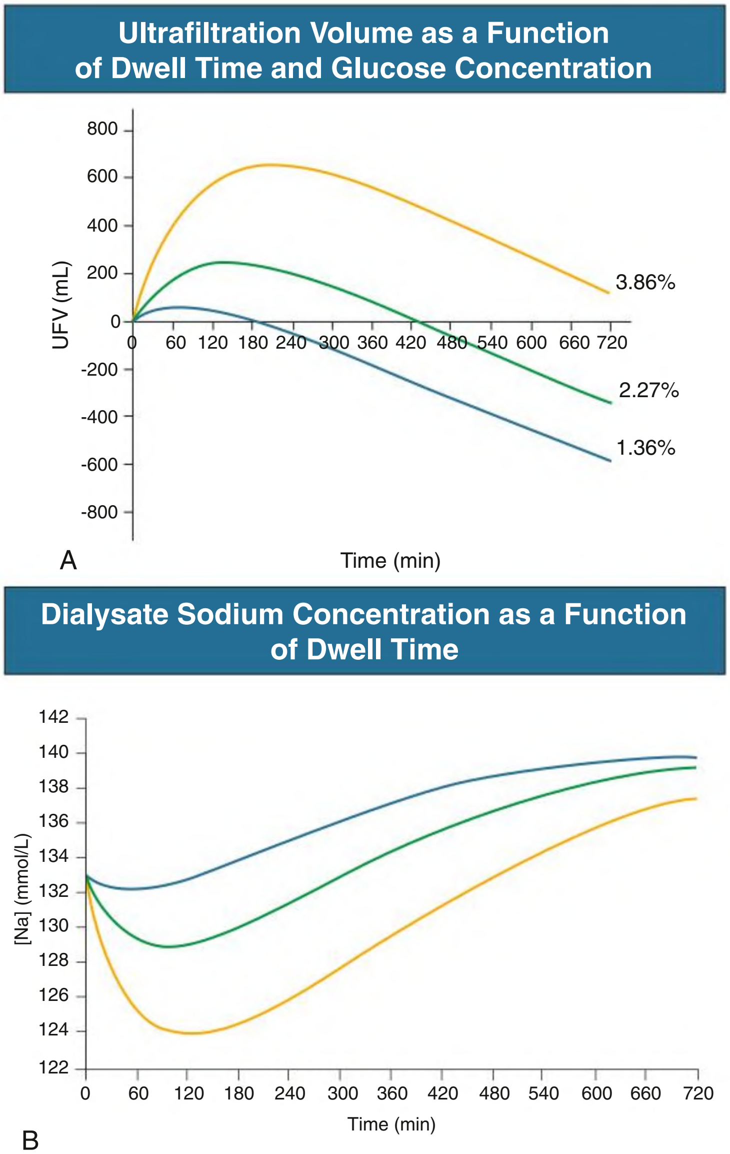

Fig. 101.3 (A) Ultrafiltration as a function of dwell time. Net ultrafiltration volume (UFV) as a function of dwell time for 3.86% *(yellow line)*, 2.27% *(green line)*, and 1.36% *(blue line)* glucose, computer simulated using the three-pore model of peritoneal transport. (B) Dialysate sodium as a function of dwell time. Dialysate sodium as a function of dwell time for 3.86% *(yellow line)*, 2.27% *(green line)*, and 1.36% *(blue line)* glucose, computer simulated according to the three-pore model of peritoneal transport.

123 mmol/L within 90 minutes, followed by a gradual increase toward the serum sodium concentration. On the other hand, icodextrin (see later) has an average molecular weight between 13 and 19 kDa with a higher osmotic efficiency (σ ~0.5) compared with glucose across the small pores and is inefficient across aquaporins.[15] This inefficiency of icodextrin across aquaporins results in minimal UF across these channels and explains why this solution does not result in sodium sieving (Fig. 101.5B). The important clinical implication is that the volume of ultrafiltrate induced by icodextrin will contain more sodium than the volume induced by a dextrose-based solution.

TYPES OF PERITONEAL DIALYSIS FLUID

PD fluid is usually packaged in clear polyvinyl chloride bags containing lactate-buffered, tightly regulated electrolyte levels without much variation between manufacturers. The solutions are potassium free, which allows for easier clearance of potassium down a concentration gradient. The primary osmotic agents used in PD fluids are glucose, amino acids, and icodextrin. Traditionally, solutions were prepared using a lactate-buffer instead of bicarbonate, as Ca^{2+} and Mg^{2+} may precipitate to form calcium carbonate and magnesium carbonate, respectively, at an elevated pH during storage. Commercially available solutions now include multichambered PD delivery systems that replace lactate with bicarbonate. In dialysis bags with multiple chambers, the glucose is stored under conditions of low pH (the buffer is in another chamber that is admixed before inflow), and the production of glucose degradation productions during sterilization or storage is delayed by this low pH (see biocompatible solutions).[16]

Glucose-Containing Solutions

Glucose-based solutions are the primary osmotic agents used in the management of PD. Options include solutions of low concentration (1.36% or 1.5%), intermediate concentration (2.27% or 2.5%), and high concentration (3.86% or 4.5%). The approximates in osmolarities

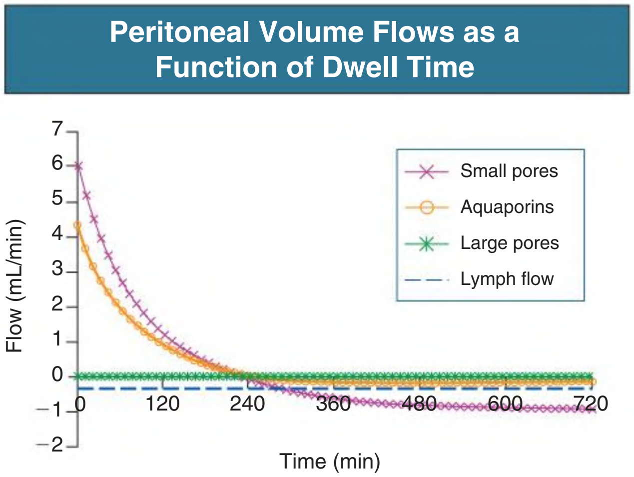

Fig. 101.4 Peritoneal Volume Flows as a Function of Dwell Time. Peritoneal volume flows as a function of dwell time for 3.86% glucose partitioned among aquaporins, small pores, large pores, and lymphatic absorption. The small-pore volume flow is initially approximately 60% of total volume flow and becomes negative after peak time (220 min). The aquaporin-mediated water flow becomes slightly negative after approximately 250 minutes. The large-pore volume flow is negligible and remains constant throughout the dwell, as does lymphatic absorption (0.3 mL/min). (Modified from Venturoli D, Rippe B. Validation by computer simulation of two indirect methods for quantification of free water transport in peritoneal dialysis. *Perit Dial Int.* 2005;25[1]:77–84.)

are 345 mOsm/L (low concentration), 395 mOsm/L (intermediate concentration), and 484 mOsm/L (high concentration). A very-low-concentration solution (0.5%) is less readily available and can be offered to patients with hypotension or intravascular volume contraction. The 0.5% solution leads to water and electrolyte absorption from dialysate to plasma and can sometimes be used to avoid intravenous fluid resuscitation.

There are, however, important drawbacks associated with the chronic use of glucose-based solutions. These include hyperglycemia, dyslipidemia as a result of continuous glucose absorption, and declining osmotic gradient ($\sigma = 0.03$) over time (Fig. 101.3A). Additionally, heat sterilization of glucose can lead to the generation of glucose degradation products (GDPs) (~5.5), which could have detrimental effects on the peritoneal membrane.[17] These reactive carbonyl compounds have also been shown to have toxic effects on various cells in vitro and in vivo.[18,19]

Non–Glucose-Based Solutions

Given the highlighted drawbacks of glucose-based solutions, alternative osmotic agents including amino acids and icodextrin are available. Icodextrin is an isoosmolar glucose polymer (~280 mOsm/L) that is manufactured as a 7.5% solution with the same electrolyte constituents as glucose-based solutions. It has a molecular weight averaging between 13 and 19 kDa that generates UF by its oncotic effect.[20] The large molecules greatly enhance the osmotic efficiency of this solution ($\sigma = 0.5$) compared with glucose solutions, leading to more sustained UF over a 12-hour period (Fig. 101.5A), and so icodextrin is preferred for long dwell exchanges (daytime dwell for patients on APD and overnight dwell for CAPD). Although icodextrin can be slowly absorbed into the plasma and metabolized to produce maltose, there have been no reported events of maltose toxicity. However, maltose can interact with glucose dehydrogenase pyrroloquinoline quinone assays used for capillary glucose measurement, leading to falsely elevated results for glucose. Therefore, alternative strategies for blood glucose monitoring must be considered in patients being treated with icodextrin, or it must be ensured that the glucometer being used is compatible with use of icodextrin and does not read out maltose as glucose.[21]

Amino acid–based solutions are available as a 1.1% amino acid mixture that has a similar osmolality to 1.36% glucose solution.[22] Like icodextrin, these solutions have the theoretical benefit of reducing the overall glucose exposure for the peritoneal membrane. Amino acid–based solutions are used primarily to enhance nutrition, but the results have been disappointing.[23,24] These solutions are limited to one exchange per day, as the absorbed amino acids can lead to acidosis and increase plasma urea levels. The access to amino acid–based solutions is now limited to European countries and a few other countries in compassionate use programs.

"Biocompatible" Solutions

The previously mentioned multicompartment systems are mixed just before instillation, thus addressing two potential limitations of traditional single-compartment lactate-buffered solutions. First, mixing just before instillation allows bicarbonate to be used as the buffer instead of lactate. The pH of lactate-buffered solutions is acidic (~5.5), whereas the pH of bicarbonate-buffered solutions is approximately neutral (~7.4),[19,25-27] which may reduce infusion pain while still correcting metabolic acidosis.[28] Second, the use of two compartments allows glucose to be stored at very low pH during heat sterilization, which protects it from caramelization and GDP formation. Experimental and observational data suggest that these solutions should reduce the patient's exposure to GDPs and that this should lead to clinical benefit.

However, clinical data has shown conflicting results in terms of overall benefits with the use of these "biocompatible" solutions. For example, one study demonstrated an improvement in RKF among patients receiving PD solutions with low GDP concentration.[25] However, more recent randomized trials have not supported such findings.[28,29] The balANZ study was a small randomized controlled trial (RCT) that suggested biocompatible solutions had lower incidence of peritonitis compared with conventional solutions.[30] There have been other small RCTs with conflicting results with regards to peritonitis and RKF. Systematic reviews have also shown inconclusive findings, and as a result the overall benefit of biocompatible solutions remains unclear.[26,27]

In commercial PD solutions the concentration of Na^+, Cl^-, Ca^{2+}, and Mg^{2+} are provided at levels similar to their respective plasma concentrations.[11] As a result, the diffusive capacity of these electrolytes is greatly reduced, and their transport across the peritoneal membrane is mostly dependent on convective transport (with UF). Computer simulations demonstrate that for an average patient with a plasma concentration of 140 mmol/L, approximately 10 mmol of Na^+ is removed with every 100 mL of UF volume produced.[11] Na^+ concentration in the solution is usually 132 to 134 mmol/L, as higher concentrations would reduce the diffusive gradient for sodium removal from the plasma.

Similarly, Ca^{2+} transport is primarily driven by convective transport, a UF-dependent process. The concentration of Ca^{2+} in PD solutions usually ranges from 1.25 to 1.75 mmol/L. In reducing the risk of hypercalcemia in PD patients who commonly use calcium-based phosphate binders, a calcium concentration of 1.25 mmol/L in the dialysate is recommended to achieve a "calcium neutral" state,[11] in which calcium is neither lost nor gained during a dwell. In commercial solutions, the concentration of Ca^{2+} does not vary with glucose concentrations, and so solutions with 1.25 mmol/L of Ca^{2+} may be preferable for patients treated with calcium-based phosphate binders to reduce the risk of hypercalcemia. Net peritoneal calcium removal can be obtained by intermediate- and high-concentration solutions with a Ca^{2+} concentration of 1.25 mmol/L,

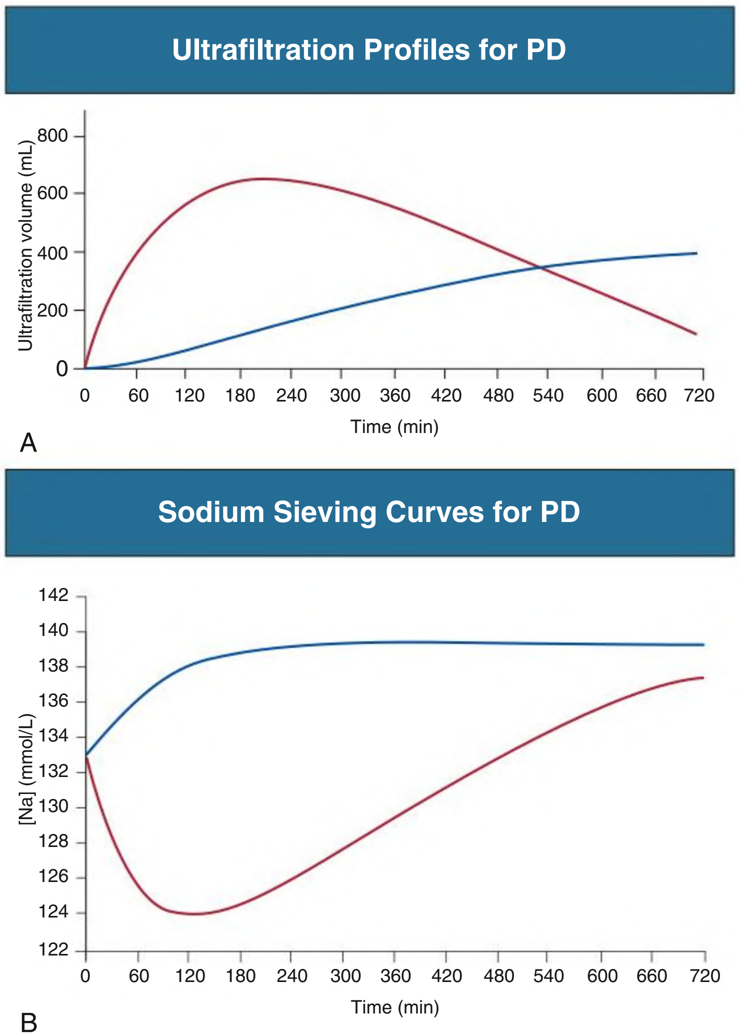

Fig. 101.5 (A) Ultrafiltration (UF) profiles for peritoneal dialysis (PD). UF profile for 7.5% icodextrin *(blue line),* computer simulated according to the three-pore model in an average patient who is not naive to icodextrin, in comparison with the computer-simulated UF curve for 3.86% glucose *(red line).* (B) Sodium-sieving curves for PD. 7.5% icodextrin *(blue line)* and 3.86% glucose *(red line)* (see Fig. 101.3). *Na,* Sodium. (A, From Rippe B, Levin L. Computer simulations of ultrafiltration profiles for an icodextrin-based peritoneal fluid in CAPD. *Kidney Int.* 2000;57[6]:2546–2556.)

as convective transport results in 0.1 mmol of Ca^{2+} for every 100 mL of UF.[11]

The other important divalent cation in PD solutions is magnesium. The concentration of Mg^{2+} in manufactured solutions ranges from 0.25 to 0.75 mmol/L. In 1.5% (1.36%) glucose solutions, the concentration of Mg^{2+} at 0.25 mmol/L would maintain a net neutral magnesium level during the dwell,[31] whereas net losses of Mg^{2+} would be seen with PD fluid with higher glucose concentrations due to larger convective flux driven by UF.

PERITONEAL ACCESS

Peritoneal Dialysis Catheters

A well-functioning PD catheter is critical for sustainable treatment with PD, because approximately 20% of PD withdrawal is due to catheter dysfunction.[32] The Tenckhoff catheter is a Silastic tube with numerous holes along the intraperitoneal segment.[33] Modern catheters typically have two Dacron (polyester) cuffs that limit catheter migration and reduce risks of pericatheter leaks and infection. The deep cuff is best placed in the rectus muscle of the abdominal wall, which allows the tissue ingrowth to encapsulate the catheter, reducing the risk of migration. On the other hand, the superficial cuff is best placed about 2 to 4 cm from the exit site. This functions as a barrier against the entry of bacteria into the subcutaneous tunnel.[34]

Catheters are available in coiled-tip and straight-tip types, with further variation existing in some catheters whereby there is a preformed bend in the intercuff segment called a “swan neck” (Fig. 101.6). Reported benefits of less inflow pain associated with coiled-tip variants due to better dispersion of dialysate fluid rather than a rapid jet flow with straight-tip variants remain hypothetical. Numerous studies comparing outcome and complication rates between the two catheter-tip variants have produced conflicting results,[35-37] and thus either is acceptable.

Insertion of Peritoneal Dialysis Catheters

PD catheters should be inserted under sterile conditions by a skilled operator. Laparoscopic insertion is the preferred method, as it allows the operator to incorporate interventions to enhance the overall success rate of functioning catheter. Operators can insert catheters with laparoscopically guided rectus sheath tunneling directed toward the pelvis,

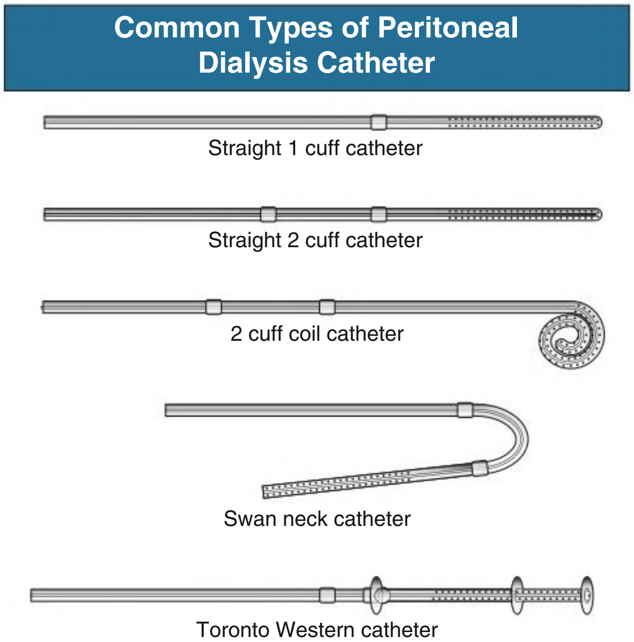

Fig. 101.6 Various types of peritoneal dialysis catheter.

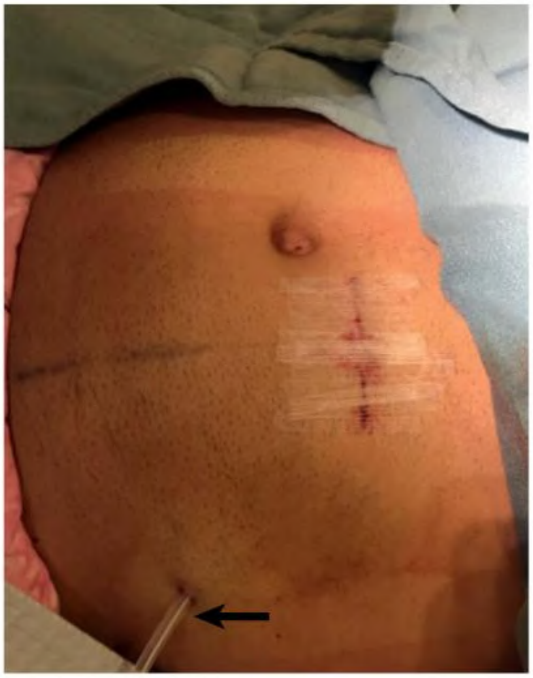

Fig. 101.7 Peritoneal Dialysis Catheter Inserted at the Bedside. Note the insertion in the left paramedian portion of the abdomen through which the catheter was inserted using a Seldinger technique *(arrow)*.

which reduces the likelihood of catheter tip migration. Furthermore, laparoscopic insertion allows operators to visualize redundant omentum that may compromise catheter function; redundancy is reduced by suturing it to a nearby organ or upon itself in a procedure referred to as an omentopexy. Alternatively, percutaneous insertion using needle with guidewire and the assistance of ultrasound is a technique that can be performed at the bedside (Fig. 101.7). This technique uses a modified Seldinger technique and is useful in low-resource settings in patients who do not have abdominal hernias and previous surgical abdominal history.[34]

Before catheter insertion, patients should undergo a complete abdominal assessment to identify any prior surgical scars or hernias and to plan for the catheter exit site. It is important that the exit site be planned with the patient sitting rather than when lying on the table so as to account for any pannus that could obscure the site. Antibiotic prophylaxis should be given to patients before catheter insertion to reduce the risk of peritonitis.[38] A single preoperative dose of a first- or second-generation cephalosporin is recommended depending on local resistance patterns and patient allergic history.

PERITONEAL DIALYSIS TECHNIQUES

Historically, transporter status guided prescriptive strategies for PD patients. For example, high transporters would have a rapid dissipation of their glucose gradient and theoretically were deemed best suited for APD. On the other hand, patients with low transporter status were deemed best suited for CAPD, allowing for dwells of longer duration in effort to maximize diffusive transport. Although theoretically these prescriptive strategies may maximize the efficiency of peritoneal transport, it is not a practical approach used in everyday practice. PD prescriptions should be individualized according to patient priorities and lifestyle rather than relying on transport kinetics. For instance, patients who have undergone recent abdominal surgery or have a history of hernias may be better suited for cycler-based APD therapy. This technique provides rapid exchanges for patients while maintaining adequate dialysis without significantly increasing IPP, in turn reducing the risk of leaks and other complications. Additionally, patients with daytime employment may find it challenging to perform exchanges during the day. Nightly APD followed by a long day-dwell allows flexibility in the daytime schedule for patients. Automated cyclers have evolved with newer technological features that allow for remote monitoring of inflow time, dwell time, and drain time along with drain volumes. Most APD machines are equipped with warmers that can warm fluid before inflow into the peritoneal cavity, reducing the episodes of discomfort and shivering that may occur with room temperature fluid. In certain patients, low drain alarms may occur on the cycler, which affects the efficiency of an exchange. This can be overcome by increasing the residual intraperitoneal volume and proportionally decreasing subsequent fill volumes in a process called tidal peritoneal dialysis (TPD). For example, 70% TPD implies that, for a 2-L inflow volume, 1.4 L would be drained out, and the next inflow volume would also be 1.4 L, leaving a residual intraperitoneal volume of about 0.6 L repeated over the cycles until the final drain. TPD was initially developed to enhance the diffusive clearance of solutes, but results have been disappointing; TPD is now used primarily for the management of low drain alarms on the cycler and for those experiencing pain during the inflow and/or drain phase of the exchange.

Alternatively, patients can be managed with CAPD using a volume of 1.5 to 2.5 L of fluid with up to five exchanges daily depending on RKF. CAPD day-dwell times can range from 4 to 6 hours before dissipation of the glucose osmotic gradient. The overnight dwell for CAPD can last for a period of 10 to 12 hours, where icodextrin or a high-concentration glucose-containing solution can be used to sustain an osmotic gradient.

PD is also dependent on a functioning connection system that is safe and simple to use. There are three major types of connection systems used in PD. They include the “standard” or straight connecting system, the Y connecting system, and the double bag system. The standard connecting system has fallen out a favor due to higher rates of peritonitis and is now rarely used. The Y connecting system uses a Y-shaped tube that attaches to an extension tube from the patient’s catheter; the two other limbs of the system are connected to unused PD fluid and a sterile empty drain bag. This system uses a principle called "flush before fill," which is likely the underlying reason that peritonitis rates are lower with this system. Intraluminal microorganisms are flushed into the drain bag, avoiding intraperitoneal contamination.[39] The original Y system developed in the 1980s has since evolved to the double-bag system, which is the most widely used connection system.[40] In this system the solution bag and drain bags are already attached to the two limbs of the Y connector; patients now have to make one simple connection to extension tubing rather than three with the original system.[40] The double-bag system is flushed for about 5 seconds and then a "frangible" (breakable) pin within the tubing is broken, allowing the spent dialysate from the peritoneal cavity to fill in the drain bag for up to 20 minutes. The limb to the drain bag is then clamped and the limb from the PD solution is opened to fill the peritoneal cavity. The time for exchange, accounting for the instillation and drainage, should not exceed 30 minutes provided that the PD catheter is functional. The initial volume (the first 1.6–1.8 L) from the peritoneal cavity usually drains rapidly, but this rate decreases as the residual intraperitoneal volume (sump volume) falls to less than 300 mL.

PERITONEAL SOLUTE TRANSPORT AND ULTRAFILTRATION

Small-Solute Removal

Most solutes typically removed by normal functioning kidneys are primarily removed by diffusive clearance in PD. The two methods commonly used in PD to assess solute clearance are the urea clearance normalized to total body water, expressed as Kt/V_{urea} (where *Kt* is the weekly clearance and V_{urea} is the volume of distribution of urea), and the peritoneal creatinine clearance standardized to body surface area (1.73 m^2).

In determining Kt, daily clearance must first be established by the measurement of urea and creatinine with PD in using the dialysate-plasma concentration ratios (D/P). The D/P_{urea} and $D/P_{creatinine}$ for either solute is done by measuring the concentration of the solute in the "used" dialysate and that in the plasma multiplied by the daily drain volume to provide the 24-hour clearance, providing the daily clearance. The weekly clearances can then be calculated by multiplying these values by 7.

Despite more contemporary studies showing little association between small-solute kinetics and patient outcomes, Kt/V_{urea} is still used as a marker of adequacy.[41,42] However, some centers use peritoneal creatinine clearance as a marker of solute removal given the data for Kt/V_{urea} and the imprecision in determining V.

Large-Solute Removal

Large solutes are removed more slowly compared with small solutes with PD via convective clearance. Although surrogate markers for large-solute clearance can be measured using albumin, immunoglobulins, and α_2-macroglobulin, this is not common practice. These solutes move across the large pores very slowly, such that equilibration in the dialysate with the plasma does not occur with the typical dwell time that is used for PD. Nevertheless, removal of large solutes is time dependent, and so the long dwell is important for this process.

Ultrafiltration

The volume of UF achieved with PD can vary significantly from day to day. This variability stems from changes in the tonicity of the solution used and patient positioning along with common factors affecting drainage such as constipation, which may lead to elevations in residual intraperitoneal volume. Collecting fluid volumes over several days and using the average value improves the accuracy of estimated daily UF

BOX 101.1 Peritoneal Equilibration Test

1. Two liters (warm) 2.27% fluid are instilled for 10 min with the patient supine and rolling from side to side every 2 min.
2. Exactly 10 min after start of the infusion, 200 mL is drained into the bag. Draw 5 mL (discard); the next 5 mL is taken for creatinine and glucose determination.
3. After 2 h, new samples are collected as in step 2.
4. After 4 h exactly, collect drainage over 20 min. Note total bag weight. Subtract empty bag weight. Take samples (after mixing) for creatinine and glucose.
5. Glucose D/D_0 (ratio of dialysate glucose at 4 h and at time zero) and creatinine D/P (ratio of dialysate and serum creatinine at 4 h) are plotted versus time (see Fig. 101.8). Record the total drain volume.

The night bag (8–12 h) must be 1.36% or 2.27% glucose, drained for 20 min with patient sitting upright.

volume. If UF volumes remain low, then patients should be evaluated for ultrafiltration failure (UFF) by using a high-concentration solution (3.86%/4.25%). A UF volume of less than 400 mL after a 4-hour dwell with high-concentration solution is consistent with UFF. Additionally, UF volumes can also be evaluated using the PET (see later). In determining intraperitoneal volume as a function of time, iodine-125 (^{125}I)–human serum albumin (HSA) can be used as a more accurate measurement of fluid kinetics in PD,[43] although this is rarely done in clinical practice.

ASSESSING PERITONEAL MEMBRANE FUNCTION

Peritoneal Equilibration Test

The peritoneal equilibration test (PET) provides an estimation of small-solute transport and UF capacity[44] by using equilibration ratios between the dialysate and plasma (D/P) for various solutes across the peritoneal membrane. These include urea (D/P_{urea}), creatinine ($D/P_{creatinine}$), and sodium (D/P_{sodium}). The procedure is summarized in Box 101.1 and traditionally involves the instillation of 2 L of an intermediate (2.27% or 2.5%) glucose-based solution; more recently, some centers have used 4.25% solutions. As the fluid is instilled over a period of about 10 minutes, the patient is encouraged to roll from side to side while in a supine position. This allows the infused fluid to be well distributed in the peritoneal cavity, particularly in the paracolic gutters. After the fill is completed, dialysate samples are taken three times: at time zero, 2 hours postfill, and 4 hours postfill. At the end of the 4-hour dwell, the dialysate drained is measured to calculate the total UF. During the test, the concentrations of glucose, creatinine, urea, and sodium are noted in three dialysate samples and the plasma. These values are used to determine D/P ratios; glucose is expressed differently as the glucose concentration in the 4-hour dialysate relative to that at time zero ($D/D_{0\ glucose}$). The $D/P_{creatinine}$ and $D/D_{0\ glucose}$ values are used to categorize patients into one of four different transporter categories: slow, slow average, fast average, or fast (Fig. 101.8). Although transport characteristics may change with time due to alterations in peritoneal membrane, nearly 70% of patients have stable transport status 1 year post-PET and about 50% of patients at 2 years post-PET.

Alternatives to the traditional PET include the mini-PET and the double-mini PET. The mini-PET provides a quicker method to assess small-solute and free water transport. It uses a high-concentration (3.86% or 4.25%) glucose-based solution that is infused intraperitoneally and dwells for 1 hour.[45] The concentration of sodium is measured in the drained dialysate sample; free water transport across the

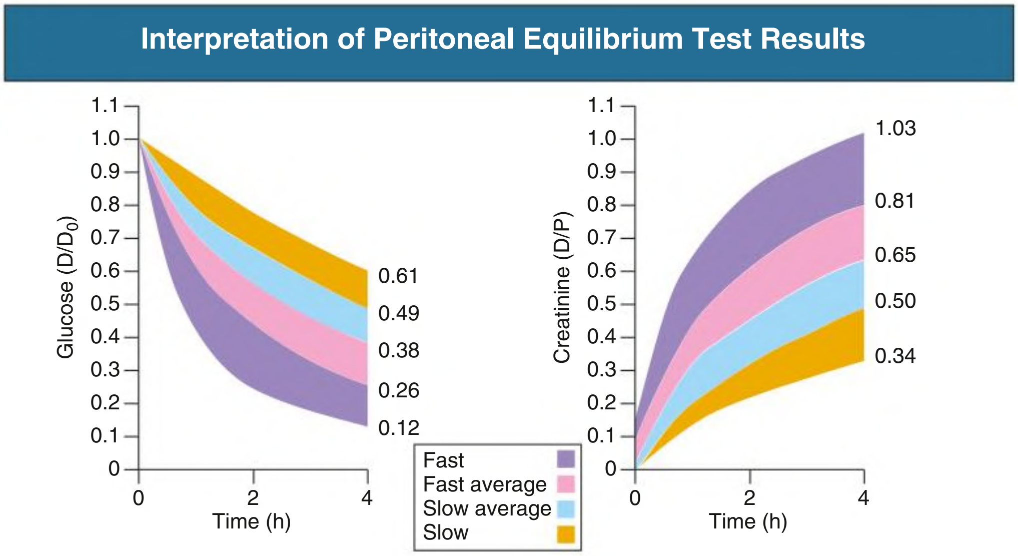

Fig. 101.8 **Interpretation of Peritoneal Equilibration Test Results.** Changes in solute concentration during a peritoneal equilibration test allow classification into different transport types. *D/D_0*, Ratio of dialysate glucose at 4 hours and at time zero) *D/P*, ratio of dialysate and serum creatinine at 4 hours. (Modified from Twardowski ZJ, Nolph KD, Khanna R. Limitations of the peritoneal equilibration test. *Nephrol Dial Transplant.* 1995;10[11]:2160–2161.)

membrane is determined indirectly. The double-mini PET test is used to evaluate the free transport of water and osmotic conductance of glucose (OC_G) across differing concentrations of solutions. The test is conducted by performing two consecutive mini-PETs[46]: the first uses a low-concentration (1.36% or 1.5%) glucose, and the second uses a high-concentration (3.86% or 4.25%) glucose solution. The double-mini PET can be used to calculate the OC_G without computer modeling by using the differences in drain volume between the low- and high-concentration solutions. In theory, it could be used to identify early signs of UFF in PD patients, but further data are needed.

It isn't clear that the PET has to be routinely performed in all PD patients. As discussed earlier, the transport status has historically been used to dictate the PD prescription, but more recently the choice of CAPD versus cycler is primarily driven by patient preference and finances. We no longer routinely perform the PET in our PD patients.

If a PET is done, this should occur no sooner than a month after catheter insertion, as the inflammation from the procedure can affect transport status. Similarly, a PET should not be done in a patient with current or recent peritonitis.

DOSE AND ADEQUACY OF PERITONEAL DIALYSIS

The use of absolute targets for Kt/V_{urea} to determine the adequacy of PD remains controversial. Historically, clinical practice guidelines recommended weekly Kt/V_{urea} of 2.0, but recent studies have failed to demonstrate a clinical benefit of Kt/V_{urea} values of greater than 1.7. Rather, a better survival advantage is associated with high RKF compared with a higher peritoneal solute clearance with higher Kt/V_{urea}.[41] Based on this emerging data, current clinical practice guidelines have evolved and now recommend that patients with a RKF of more than 100 mL/day have a minimal delivered total (peritoneal and renal) dose of small-solute clearance with a weekly Kt/V_{urea} of 1.7.[47]

Residual Kidney Function

RKF and its preservation has been associated with numerous benefits, including lower relative risk of death in dialysis patients.[41,48] Observational data has demonstrated that RKF is better preserved in patients on PD compared with HD.[49] RKF can easily be assessed by performing a 24-hour urine collection with assessment of the urea and creatinine concentrations along with the total urine volume. Because renal creatinine clearance may result in an overestimation of glomerular filtration rate (GFR) due to its tubular secretion, GFR can be more accurately estimated by averaging the renal creatinine and urea clearances respectively. However, if a 24-hour urine collection is impractical, the serum creatinine can be used as a surrogate marker of RKF. If serum creatinine is stable in the absence of changes in muscle mass or recent changes to the PD prescription, it can be assumed that RKF is preserved.[4]

PRINCIPLES AND PRACTICE OF INCREMENTAL DIALYSIS

The adoption of incremental PD has numerous benefits to patients and the health care system. It allows patients time to become comfortable with the therapy, and the prescription is enhanced to promote appropriate small-solute clearance if there is a decline in RKF. Simply put, as RKF declines, the dose of PD will need to increase. Longitudinal observational data has demonstrated that in comparison to standard PD prescriptions, survival rates in PD patients were similar, but the rates of hospitalizations and cost to the health system were much less in the incremental group. Incremental PD also reduces the reliance on Kt/V_{urea} and redirects more attention to clinical assessments and other biochemical data to determine adequacy. Incremental PD has some disadvantages that include a reluctance of patients to increase the dialysis dose when a decrease in RKF suggests that this is necessary. Delays in adjusting the PD prescription to achieve effective solute clearance can result in uremia in patients. Therefore, patients and caregivers should be informed about the principles of incremental PD before starting this type of treatment so that prescription changes are better accepted when they are warranted.

However, in our experience, the RKF does not usually suddenly decline in a stable outpatient unless there has been some notable intercurrent event. Therefore, the concern that a patient on an incremental prescription may suddenly become underdialyzed due to a sudden drop in RKF is unwarranted.

Prescribing Incremental Peritoneal Dialysis

In adopting the principles of incremental PD, clinicians should individualize prescriptions to patient needs while focusing on clinical symptoms in conjunction with laboratory investigations. For example, a patient who prefers to do manual exchange therapy can be started on CAPD with 1 to 2 exchanges per day at 1.5-L fill volumes with each exchange. Depending on patient comfort, fill volumes can be increased incrementally up to maximum of 2.5 L with each exchange. If a patient needs more solute clearance, then additional daytime exchanges can be incorporated if a maximal tolerated fill volume has been achieved. On the other hand, for patients receiving cycler-based therapy, nocturnal intermittent peritoneal dialysis (NIPD) is an initial preferred option. NIPD may be more attractive to patients with reduced flexibility in daytime schedules and can be provided initially with two to three exchanges overnight with volumes of 1.5 L with each exchange. If there is a gradual decline in RKF, then the fill volumes could be increased in 20% to 30% increments up to a maximum patient tolerance, or a value not exceeding 2.5 L, in order to improve solute clearance. If more solute clearance is still needed, then the number of exchanges on the cycler can gradually be increased followed by the addition of a daytime dwell.

Fluid Balance

Patients on PD have significant day-to-day variation in their fluid balance, which requires changes in the PD prescription and diuretic dosing in those with significant RKF. Maintaining euvolemia is an important predictor of outcomes in PD patients, but hypervolemia is quite common in this population.[50] Observational data have revealed that among 639 prevalent patients who had been on PD for at least 2 years, more than 70% had hypervolemia.[51] The reasons for hypervolemia are multifactorial and include loss of RKF, increased dietary intake of salt and fluid, nonadherence to dialysis, and inadequate prescription (low-concentration solution leading to rapid dissipation of osmotic gradient in a fast transporter). Hypervolemia can exacerbate hypertension in PD patients and may lead to left ventricular hypertrophy. As patients remain on PD for longer periods, their antihypertensive regimen (number of agents and their doses) often increases over time. In patients with hypervolemia, clinical assessment (in conjunction with a mini-PET or double mini-PET test if required; see earlier) is used to determine whether UFF is the underlying cause for volume overload.

Management of Fluid Overload

In treating fluid overload, a stepwise approach should be considered for all PD patients. First, patients must be encouraged to reduce dietary intake of sodium to less than 2 g/day. Second, if RKF is more than 100 mL/day and volume overload persists, then a loop diuretic such as furosemide can be increased to 320 to 400 mg/day in divided doses[52]; a thiazide-like diuretic such as metolazone can be added to enhance urinary volumes. Increased doses of diuretics serve to promote natriuresis and do not augment renal clearance of solutes. Last, if volume overload remains despite dietary modifications and increased diuretic dosing, then volume removal can be enhanced with peritoneal UF by increased glucose concentration of PD solutions. Icodextrin can be used for longer dwells, particularly for daytime dwells with APD, as this increases UF and reduces extracellular volume.[53] The rate of UF with icodextrin is much slower compared with glucose solutions with maximal UF occurring about 10 hours into the dwell. Therefore, icodextrin should not be used for the management of acute volume overload, where hypertonic solutions with faster UF profile rates are preferred.

Nutrition

The importance of good nutrition while on PD may be neglected, leading to poor patient outcomes. PD initially leads to weight gain in patients in the absence of volume overload. This weight gain is due to increase in caloric intake from peritoneal glucose reabsorption. Peritoneal glucose reabsorption leads to daily average of 400 to 600 kcal/day, which may lead to metabolic syndrome in nearly half of all prevalent PD patients.[54] However, despite glucose reabsorption, patients may still have energy malnutrition as a result of uremic wasting syndrome. Furthermore, as RKF declines, protein-energy malnutrition and wasting increases due to impairment in protein metabolism in the presence of uremia.[55] The suggested daily energy target for PD patients is at least 35 kcal/kg/day, with about 10% to 30% of this target being derived from the glucose-based PD solutions. Considering protein losses with PD (5–7 g/day, 80% of which is albumin), the recommended dietary daily protein target should be more than 1.2 g protein/kg/day.

Acknowledgment

We thank Dr. Bengt Rippe (1950–2016) for his contributions to the previous edition. Dr. Rippe was a trailblazer in the field of physiology, with his research findings being the bedrock of many applied principles of PD in today's practice.

SELF-ASSESSMENT QUESTIONS

1. In PD, the fraction of AQP-mediated water flow is much higher for glucose than for ICO as osmotic agent. Why?
   A. Glucose will induce more initial vasodilation and recruitment of exchange vessel surface area than ICO.
   B. The glucose molecules are smaller than the ICO molecules and therefore are relatively "inefficient" as osmotic agents in small pores but relatively more efficient across AQP.
   C. Glucose, but not ICO, will regularly induce an increase in the number of AQPs.
   D. Glucose, but not ICO, can increase the peritoneal capillary filtration coefficient.
   E. Oncotic agents such as ICO cannot pull fluid across AQP at all.
2. Which PD catheter has a superior clinical outcome?
   A. The Toronto Western catheter
   B. The Swan neck catheter
   C. The straight one-cuff (original) Tenckhoff catheter
   D. The straight two-cuff (original) Tenckhoff catheter
   E. None of these catheters shows significant clinical superiority
3. Long-term changes in peritoneal transport parameters usually involve:
   A. increases in PET_{creat} and marked increases in the peritoneal UF coefficient.
   B. reductions in PET_{creat} and increases in the peritoneal UF coefficient.
   C. increases in PET_{creat} combined with no or only moderate increases in the UF coefficient.
   D. increases in large-solute transport together with increases in UF coefficient.
   E. increases in large-solute transport combined with reductions in the UF coefficient.
4. Glucose-based dialysis fluids for PD that are low in GDPs have which of the following advantages?
   A. They prevent peritoneal fibrosis and encapsulating peritoneal sclerosis.
   B. They preserve residual renal function.
   C. They prevent increases in small-solute transport over treatment time.

D. They produce higher concentrations of the dialysate effluent marker CA-125.
E. They increase peritoneal UF capacity.

5. Which of the following is true about icodextrin as a high molecular weight osmotic agent?
A. ICO is monodispersed.
B. ICO is polydispersed, but with 100% of the molecules being larger than 3 kDa.
C. ICO is polydispersed, but with 80% of the molecules being larger than 3 kDa.
D. ICO is polydispersed, but with 50% of the molecules being larger than 3 kDa.
E. ICO is polydispersed, but with 30% of the molecules being larger than 3 kDa.

6. Net fluid reabsorption from the peritoneal cavity occurs via lymphatic reabsorption plus backfiltration through:
A. small pores.
B. AQP-1.
C. large pores.
D. larges pores + AQP-1.
E. small pores + large pores.

REFERENCES

1. Li PK, Chow KM, Van de Luijtgaarden MW, et al. Changes in the worldwide epidemiology of peritoneal dialysis. *Nat Rev Nephrol*. 2017;13(2):90–103.
2. Lysaght MJ, Vonesh EF, Gotch F, et al. The influence of dialysis treatment modality on the decline of remaining renal function. *ASAIO (Am Soc Artif Intern Organs) Trans*. 1991;37(4):598–604.
3. Tang M, Li T, Liu H. A comparison of transplant outcomes in peritoneal and hemodialysis patients: a meta-analysis. *Blood Purif*. 2016;42(2):170–176.
4. Auguste BL, Bargman JM. Incremental peritoneal dialysis: new ideas about an old approach. *Semin Dial*. 2018;31(5):445–448.
5. Auguste BL, Agarwal A, Ibrahim AZ, et al. A single-center retrospective study on the initiation of peritoneal dialysis in patients with cardiorenal syndrome and subsequent hospitalizations. *Can J Kidney Health Dis*. 2020;7:2054358120979239.
6. Rippe B. A three-pore model of peritoneal transport. *Perit Dial Int*. 1993;13(suppl 2):S35–S38.
7. Ni J, Verbavatz JM, Rippe A, et al. Aquaporin-1 plays an essential role in water permeability and ultrafiltration during peritoneal dialysis. *Kidney Int*. 2006;69(9):1518–1525.
8. Solass W, Horvath P, Struller F, et al. Functional vascular anatomy of the peritoneum in health and disease. *Pleura Peritoneum*. 2016;1(3):145–158.
9. Leypoldt JK. Evaluation of peritoneal membrane permeability. *Adv Ren Replace Ther*. 1995;2(3):265–273.
10. Fischbach M, Dheu C. Hydrostatic intraperitoneal pressure: an objective tool for analyzing individual tolerance of intraperitoneal volume. *Perit Dial Int*. 2005;25(4):338–339.
11. Rippe B, Venturoli D, Simonsen O, de Arteaga J. Fluid and electrolyte transport across the peritoneal membrane during CAPD according to the three-pore model. *Perit Dial Int*. 2004;24(1):10–27.
12. Imholz AL, Koomen GC, Voorn WJ, Struijk DG, Arisz L, Krediet RT. Day-to-day variability of fluid and solute transport in upright and recumbent positions during CAPD. *Nephrol Dial Transplant*. 1998;13(1):146–153.
13. Rippe B, Stelin G, Haraldsson B. Computer simulations of peritoneal fluid transport in CAPD. *Kidney Int*. 1991;40(2):315–325.
14. Devuyst O, Rippe B. Water transport across the peritoneal membrane. *Kidney Int*. 2014;85(4):750–758.
15. Rippe B, Levin L. Computer simulations of ultrafiltration profiles for an icodextrin-based peritoneal fluid in CAPD. *Kidney Int*. 2000;57(6):2546–2556.
16. Drugs.com. Extraneal Prices, Coupons and Patient Assistance Programs. https://www.drugs.com/price-guide/extraneal#:~:text=The%20cost%20for%20Extraneal%20intraperitoneal,on%20the%20pharmacy%20you%20visit.
17. Kjellstrand P, Erixon M, Wieslander A, Linden T, Martinson E. Temperature: the single most important factor for degradation of glucose fluids during storage. *Perit Dial Int*. 2004;24(4):385–391.
18. Wieslander AP, Nordin MK, Kjellstrand PT, Boberg UC. Toxicity of peritoneal dialysis fluids on cultured fibroblasts, L-929. *Kidney Int*. 1991;40(1):77–79.
19. Rippe B, Simonsen O, Heimburger O, et al. Long-term clinical effects of a peritoneal dialysis fluid with less glucose degradation products. *Kidney Int*. 2001;59(1):348–357.
20. Mistry CD, Gokal R, Peers E. A randomized multicenter clinical trial comparing isosmolar icodextrin with hyperosmolar glucose solutions in CAPD. MIDAS Study Group. Multicenter Investigation of Icodextrin in Ambulatory Peritoneal Dialysis. *Kidney Int*. 1994;46(2):496–503.
21. Oyibo SO, Pritchard GM, McLay L, et al. Blood glucose overestimation in diabetic patients on continuous ambulatory peritoneal dialysis for end-stage renal disease. *Diabet Med*. 2002;19(8):693–696.
22. Kopple JD, Bernard D, Messana J, et al. Treatment of malnourished CAPD patients with an amino acid based dialysate. *Kidney Int*. 1995;47(4):1148–1157.
23. Asola M, Virtanen K, Nagren K, et al. Amino-acid-based peritoneal dialysis solution improves amino-acid transport into skeletal muscle. *Kidney Int Suppl*. 2008;(108):S131–S136.
24. Li FK, Chan LY, Woo JC, et al. A 3-year, prospective, randomized, controlled study on amino acid dialysate in patients on CAPD. *Am J Kidney Dis*. 2003;42(1):173–183.
25. Williams JD, Topley N, Craig KJ, et al. The Euro-Balance Trial: the effect of a new biocompatible peritoneal dialysis fluid (balance) on the peritoneal membrane. *Kidney Int*. 2004;66(1):408–418.
26. Cho Y, Johnson DW, Craig JC, Strippoli GF, Badve SV, Wiggins KJ. Biocompatible dialysis fluids for peritoneal dialysis. *Cochrane Database Syst Rev*. 2014;(3):CD007554.
27. Cho Y, Badve SV, Hawley CM, Wiggins K, Johnson DW. Biocompatible peritoneal dialysis fluids: clinical outcomes. *Int J Nephrol*. 2012;2012:812609.
28. Mactier RA, Sprosen TS, Gokal R, et al. Bicarbonate and bicarbonate/lactate peritoneal dialysis solutions for the treatment of infusion pain. *Kidney Int*. 1998;53(4):1061–1067.
29. Fan SL, Pile T, Punzalan S, Raftery MJ, Yaqoob MM. Randomized controlled study of biocompatible peritoneal dialysis solutions: effect on residual renal function. *Kidney Int*. 2008;73(2):200–206.
30. Johnson DW, Brown FG, Clarke M, et al. Effects of biocompatible versus standard fluid on peritoneal dialysis outcomes. *J Am Soc Nephrol*. 2012;23(6):1097–1107.
31. Hutchison AJ, Merchant M, Boulton HF, Hinchcliffe R, Gokal R. Calcium and magnesium mass transfer in peritoneal dialysis patients using 1.25 mmol/L calcium, 0.25 mmol/L magnesium dialysis fluid. *Perit Dial Int*. 1993;13(3):219–223.
32. Workeneh B, Guffey D, Minard CG, Mitch WE. Causes for withdrawal in an urban peritoneal dialysis program. *Int J Nephrol*. 2015;2015:652953.
33. Tenckhoff H, Schechter H. A bacteriologically safe peritoneal access device. *Trans Am Soc Artif Intern Organs*. 1968;14:181–187.
34. Crabtree JH, Chow KM. Peritoneal dialysis catheter insertion. *Semin Nephrol*. 2017;37(1):17–29.
35. Ouyang CJ, Huang FX, Yang QQ, et al. Comparing the incidence of catheter-related complications with straight and coiled Tenckhoff catheters in peritoneal dialysis patients-a single-center prospective randomized trial. *Perit Dial Int*. 2015;35(4):443–449.
36. Hagen SM, Lafranca JA, I.J JN, Dor FJ. A systematic review and meta-analysis of the influence of peritoneal dialysis catheter type on complication rate and catheter survival. *Kidney Int*. 2014;85(4):920–932.
37. Nielsen PK, Hemmingsen C, Friis SU, Ladefoged J, Olgaard K. Comparison of straight and curled Tenckhoff peritoneal dialysis catheters implanted by percutaneous technique: a prospective randomized study. *Perit Dial Int*. 1995;15(1):18–21.
38. Szeto CC, Li PK, Johnson DW, et al. ISPD catheter-related infection recommendations: 2017 update. *Perit Dial Int*. 2017;37(2):141–154.
39. Daly CD, Campbell MK, MacLeod AM, et al. Do the Y-set and double-bag systems reduce the incidence of CAPD peritonitis? A systematic review of randomized controlled trials. *Nephrol Dial Transplant*. 2001;16(2):341–347.
40. Buoncristiani U. Birth and evolution of the "Y" set. *ASAIO J*. 1996;42(1):8–11.
41. Bargman JM, Thorpe KE, Churchill DN, Group CPDS. Relative contribution of residual renal function and peritoneal clearance to adequacy of dialysis: a reanalysis of the CANUSA study. *J Am Soc Nephrol*. 2001;12(10):2158–2162.
42. Paniagua R, Amato D, Vonesh E, et al. Effects of increased peritoneal clearances on mortality rates in peritoneal dialysis: ADEMEX, a prospective, randomized, controlled trial. *J Am Soc Nephrol*. 2002;13(5):1307–1320.
43. Olszowska A, Waniewski J, Stachowska-Pietka J, Garcia-Lopez E, Lindholm B, Wankowicz Z. Long peritoneal dialysis dwells with icodextrin: kinetics of transperitoneal fluid and polyglucose transport. *Front Physiol*. 2019;10:1326.
44. Twardowski ZJ, Nolph KD, Khanna R. Limitations of the peritoneal equilibration test. *Nephrol Dial Transplant*. 1995;10(11):2160–2161.
45. La Milia V, Di Filippo S, Crepaldi M, et al. Mini-peritoneal equilibration test: a simple and fast method to assess free water and small solute transport across the peritoneal membrane. *Kidney Int*. 2005;68(2):840–846.

46. La Milia V, Limardo M, Virga G, Crepaldi M, Locatelli F. Simultaneous measurement of peritoneal glucose and free water osmotic conductances. *Kidney Int*. 2007;72(5):643–650.
47. Peritoneal Dialysis Adequacy Work G. Clinical practice guidelines for peritoneal dialysis adequacy. *Am J Kidney Dis*. 2006;48(suppl 1):S98–S129.
48. Marron B, Remon C, Perez-Fontan M, Quiros P, Ortiz A. Benefits of preserving residual renal function in peritoneal dialysis. *Kidney Int Suppl*. 2008;(108):S42–S51.
49. Adequacy of dialysis and nutrition in continuous peritoneal dialysis: association with clinical outcomes. Canada-USA (CANUSA) Peritoneal Dialysis Study Group. *J Am Soc Nephrol*. 1996;7(2):198–207.
50. Lo WK, Bargman JM, Burkart J, et al. Guideline on targets for solute and fluid removal in adult patients on chronic peritoneal dialysis. *Perit Dial Int*. 2006;26(5):520–522.
51. Van Biesen W, Williams JD, Covic AC, et al. Fluid status in peritoneal dialysis patients: the European Body Composition Monitoring (EuroBCM) study cohort. *PLoS One*. 2011;6(2):e17148.
52. Krediet RT. Preservation of residual kidney function and urine volume in patients on dialysis. *Clin J Am Soc Nephrol*. 2017;12(3):377–379.
53. Davies SJ, Woodrow G, Donovan K, et al. Icodextrin improves the fluid status of peritoneal dialysis patients: results of a double-blind randomized controlled trial. *J Am Soc Nephrol*. 2003;14(9):2338–2344.
54. Fortes PC, de Moraes TP, Mendes JG, Stinghen AE, Ribeiro SC, Pecoits-Filho R. Insulin resistance and glucose homeostasis in peritoneal dialysis. *Perit Dial Int*. 2009;29(suppl 2):S145–S148.
55. Demirci MS, Demirci C, Ozdogan O, et al. Relations between malnutrition-inflammation-atherosclerosis and volume status. The usefulness of bioimpedance analysis in peritoneal dialysis patients. *Nephrol Dial Transplant*. 2011;26(5):1708–1716.